

Abstract

We present GLIDE, a lightweight method for detecting anomalous readings in implanted orthopedic sensor telemetry. GLIDE combines a per-patient baseline with a shared population prior, and reaches an F1 of 0.87 on our held-out cohort, compared with 0.71 for the strongest baseline.

1 Introduction

Continuous telemetry from instrumented implants produces long, irregularly sampled series that are dominated by patient-specific variation. Existing anomaly detectors treat each patient independently, which discards the population signal, or pool all patients, which drowns the individual signal. We argue that both are needed, and that the trade-off can be made explicit.

2 Related Work

Prior work on physiological anomaly detection falls into two families. Reconstruction methods train an autoencoder per subject. Forecasting methods fit a predictive model and flag large residuals. Neither family reports results on implanted device telemetry, where the sampling rate is two orders of magnitude lower than in wearable settings.

3 Method

GLIDE fits a hierarchical model in which each patient's baseline is drawn from a population prior. The anomaly score for a reading is its negative log likelihood under the posterior predictive distribution. We set the shrinkage parameter λ to 0.4, chosen on the validation split.

3.1 Population Prior

The prior is estimated by pooling all training patients and fitting a Student-t distribution, which we found more robust to outliers than a Gaussian.

3.2 Per-Patient Baseline

Each patient's baseline is updated online with an exponential forgetting factor of 0.98, so that a genuine change in a patient's physiology is absorbed within roughly fifty readings rather than being flagged indefinitely.

4 Data

We use the SENTINEL cohort, which contains 1,284 patients and 4.6 million individual readings collected over 26 months across eleven surgical centres. Readings arrive roughly every four hours. Labels were assigned by two orthopedic surgeons, with disagreements resolved by a third reviewer; inter-rater agreement was 0.81 by Cohen's kappa.

5 Experiments

We compare GLIDE against three baselines: a per-patient autoencoder, a pooled gradient boosted tree, and a seasonal ARIMA residual detector. All models are trained on the same split, with 70 percent of patients used for training, 10 percent for validation and 20 percent held out for testing. The

split is by patient rather than by reading, so no patient appears in more than one partition.

5.1 Main Results

GLIDE reaches an F1 of 0.87, against 0.71 for the pooled gradient boosted tree, 0.66 for the per-patient autoencoder and 0.58 for the ARIMA residual detector. Precision is 0.84 and recall is 0.90.

5.2 Ablations

Removing the population prior drops F1 to 0.74. Removing the per-patient baseline drops it to 0.69. The two components are therefore complementary, and neither alone accounts for the improvement over the baselines.

6 Discussion

The gain comes mostly from recall on patients with fewer than two hundred readings, where a per-patient model has too little history to calibrate. On patients with long histories the three methods are within two points of each other.

7 Limitations

The SENTINEL cohort is drawn from a single device family, and all eleven centres are in one country, so the population prior may not transfer. We did not evaluate on paediatric patients. Labels reflect surgeon judgement rather than a confirmed clinical outcome.

8 Conclusion

We showed that combining a population prior with a per-patient baseline improves anomaly detection on implanted sensor telemetry, and that both components are necessary.

References

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